

# TerraManager User Manual

**Manage your collection. Document your animals. Identify enclosures using QR codes.**

This manual guides you through the most important TerraManager features — from creating your first enclosure to documenting feedings and reminders, archiving records, and backing up your data.

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# 1. Getting Started

TerraManager is a local management application for terrariums, rearing containers, enclosures, and the animals kept in them.

In addition to basic records, you can document images, husbandry parameters, feedings, feeding intervals, weights, and notes. Unique QR codes connect the enclosures created in TerraManager with the corresponding physical containers.

Animals and enclosures that are no longer active can be archived. Their data remains available and can later be viewed or restored.

The main navigation at the bottom of the screen consists of three areas:

## Enclosures

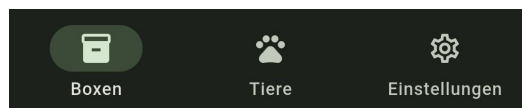
Management of terrariums, enclosures, and other containers.

## Animals

Management of your animal collection, including feedings and feeding reminders.

## Settings

QR code export, data backup, and additional settings.



*The main TerraManager navigation lets you switch directly between Enclosures, Animals, and Settings.*

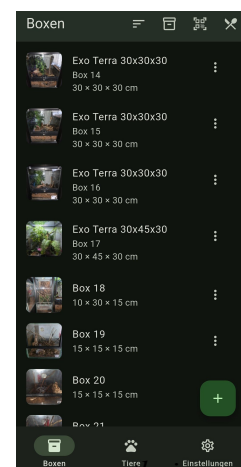
# 2. Managing Enclosures

## 2.1 Enclosure Overview

The **Enclosures** section contains all currently active containers.

Stored images make visual identification easier. The enclosure name or number also helps keep larger collections organized.

Use + to create a new enclosure at any time.



*The enclosure overview displays the terrariums and containers currently in use.*

## 2.2 Creating a New Enclosure

When creating an enclosure, you can enter the information required for later management.

This includes, in particular:

- Name
- Width, height, and depth
- Image
- Notes
- Additional information about setup or use

After saving, the enclosure receives its own enclosure ID and a unique QR identifier.

This identifier creates the connection between the physical container and its TerraManager record.

The image displays two side-by-side screenshots of the 'Neue Box' (New Box) creation interface in the TerraManager application. Both screens have a light green header with a back arrow and the title 'Neue Box'.

The left screenshot shows the initial form with the title 'Neue Box erstellen'. Below the title, it states: 'Eine dauerhafte eindeutige QR-Kennung wird automatisch erstellt. Format: TM:BOX-<UUID>'. There is a text input field for 'Boxname' with 'Optional' written below it. Below this is a large square placeholder for an image with a diagonal line and the text 'Kein Bild'. At the bottom, there is a button labeled 'Bild hinzufügen' and the text 'Abmessungen'.

The right screenshot shows the form with input fields for dimensions: 'Breite (cm)', 'Höhe (cm)', and 'Tiefe (cm)', each with 'Optional' written below it. There is also a section for 'Temperaturzonen' with 'Optional' below it, and a 'Notizen' section with 'Optional' below it. At the bottom, there is a green button labeled '+ Box erstellen'.

*When creating an enclosure, dimensions, an image, and additional information can be stored.*

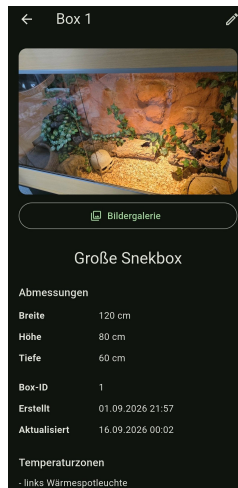
## 2.3 Enclosure Details

Tap an enclosure to open its detail view.

Depending on the information stored, the following may be displayed:

- Main image
- Name
- Dimensions
- Enclosure ID
- Creation date
- Last modified date
- Temperature information
- Notes
- Assigned animals
- QR identifier

The enclosure can be edited using the pencil icon.



*The detail view combines all information about an enclosure in one place.*

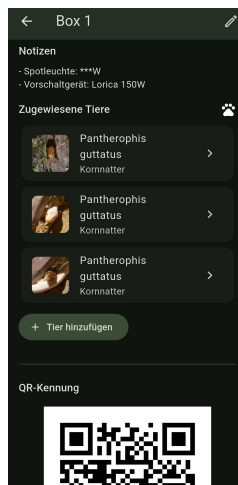
## 2.4 Assigning Animals to an Enclosure

Under **Assigned Animals**, TerraManager displays the animals currently kept in that enclosure.

An entry can be opened directly to access the corresponding animal details.

Use **Add Animal** to assign additional animals. A single enclosure can contain multiple animals.

This is intended, for example, for group housing or containers in which several animals are kept together.



*One or more animals can be assigned to an enclosure.*

# 3. Managing Animals

## 3.1 Animal Overview

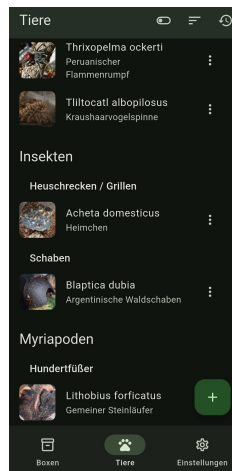
The **Animals** section contains the active animal collection.

Depending on the record, the overview may display:

- Image
- Scientific name

- Common name
- Category or grouping
- Current feeding status

Use + to create a new animal. Additional actions for existing animals are available from their menu.



*The animal overview groups the active collection and displays important information directly in the list.*

## 3.2 Categories and Subcategories

Animals can be organized using categories and subcategories.

For example, reptiles, arachnids, insects, and other animal groups can be displayed separately.

Examples:

Reptile → Snake

Arachnid → Jumping Spider

Insect → Cockroach

Myriapod → Centipede

This grouping makes navigation easier, particularly in larger collections.

## 3.3 Creating and Editing an Animal

When creating or editing an animal, extensive information can be stored.

Depending on what is known or relevant for the animal, this may include:

- Name or common name
- Scientific name
- Main image
- Assigned enclosure
- Category
- Subcategory
- Daytime temperature
- Humidity
- Date of birth
- Birth date accuracy

- Sex
- Weight
- Origin / habitat
- Nighttime temperature
- Resting or dormancy periods
- General notes

Not every field needs to be completed.

This allows TerraManager to be used both for simple collection lists and for more detailed husbandry documentation.

The image displays three sequential screenshots of the TerraManager application's 'Tier bearbeiten' (Edit Animal) screen. The first screenshot shows a photo of a zebra and basic identification fields: 'Zugehörige Box' (Criticbox - Box 34), 'Name' (Zebrajagdspinne), 'Lateinischer Name' (Viridasius spec. "sylvestrisform"), 'Kategorie' (Spinnentier), and 'Unterkategorie' (Andere Spinne). The second screenshot shows environmental and biological data fields: 'Tagestemperatur (°C)' (Min. 20.0, Max. 26.0), 'Luftfeuchtigkeit (%)' (Min. 50.0, Max. 60.0), 'Geburtsdatum' (01.01.2026), 'Genauigkeit des Geburtsdatums' (Jahr bekannt), 'Geschlecht' (Unbekannt), 'Gewicht (g)', 'Herkunft / Lebensraum' (Madagaskar), and 'Nachttemperatur (°C)'. The third screenshot shows enclosure and care notes: 'Madagaskar', 'Nachttemperatur (°C)', 'Ruhe- oder Dormanzzeiten', 'Häutungsnotizen', and 'Notizen'. At the bottom of the third screenshot are buttons for 'Speichern' (Save) and 'Tier archivieren' (Archive animal).

Basic data, enclosure assignment, and classification of an animal.

*Additional biological and husbandry-related data can be documented under **Additional Characteristics**. These extra fields allow more detailed documentation of the animal's care.*

## Approximate Birth Dates

For terrarium animals, the exact date of birth or hatching is often unknown.

TerraManager therefore stores not only the date itself but also its level of accuracy.

For example, if only the year is known, a corresponding date can be entered and the accuracy can be set to **Year Known**.

This ensures that it remains clear how precise the stored age information actually is.

## Shedding History

Individual shedding events can be recorded for an animal with a date and an optional note.

The shedding history remains chronologically traceable. Existing entries can be edited or deleted after confirmation.

A new shedding event can be added quickly from Animal Details or Edit Animal, and the complete history can be opened from the same workflows.

Adding a new shedding event does not replace earlier information. Every recorded shedding remains a separate history entry.

## 4. Animal Details

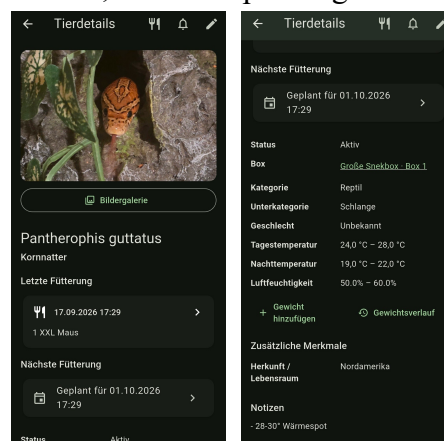
The detail view combines the information for an individual animal.

The upper section contains the main image, scientific name, and common name.

Depending on the available data, the following may also be displayed:

- Last feeding
- Next feeding
- Assigned enclosure
- Category and subcategory
- Sex
- Date of birth
- Temperature ranges
- Humidity
- Weight
- Shedding history
- Origin / habitat
- Notes

If the animal is assigned to an enclosure, the corresponding enclosure can be opened directly.



*The animal detail view combines basic data, husbandry parameters, and history functions in one place.*

## 5. Images and Gallery

### 5.1 Main Image

Enclosures and animals can be assigned images.

The respective main image is used, among other places, in overview lists and detail pages. This makes it easier to identify animals and containers visually.

Use **Change Image** to update the current main image.

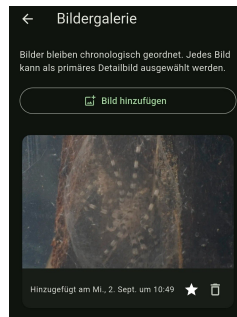
## 5.2 Image Gallery

Additional images can be stored alongside the main image.

The image gallery opens the image collection for the respective animal or enclosure.

Additional photos can be added there, and existing images can be managed.

An image can be selected as the primary image without having to delete the remaining photos.



*The image gallery allows ongoing visual documentation.*

## 6. QR Codes

### 6.1 The QR System

Each enclosure has a unique QR code.

The code can be printed and attached to the physical terrarium or container.

When the code is scanned, TerraManager immediately identifies the corresponding enclosure.

This eliminates the need to manually search for enclosure names or numbers, especially in larger collections.

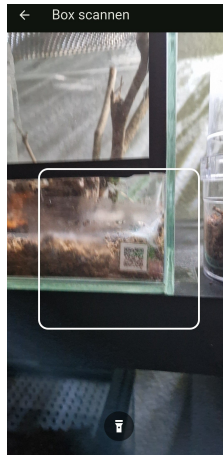


*Each enclosure receives a unique QR identifier.*

### 6.2 QR Scanner

The integrated scanner reads enclosure codes generated by TerraManager.

After successful recognition, the corresponding enclosure can be used directly for the relevant workflow.



*The QR scanner connects the physical container with its digital record.*

## 7. Exporting and Printing QR Codes

Several options for exporting enclosure QR codes are available under **Settings**. The PDF export is particularly suitable for direct printing.

### Save Enclosure QR Codes as PDF

In the dialog, the desired active enclosures can be selected.

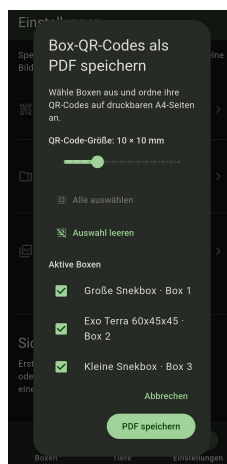
For quicker selection, **Select All** and **Clear Selection** are also available.

A slider is used to set the desired physical QR code size, for example:

**10 × 10 mm**

TerraManager then automatically arranges the selected codes on printable A4 pages.

Use **Save PDF** to create the finished file.



*QR codes can be exported together in a freely selectable size on printable A4 pages.*

In addition to PDF export, other export options are available for further individual processing of the QR codes.

## 8. Documenting Feedings

### 8.1 New Feeding

A feeding can be documented using the feeding icon for an animal.

The time as well as the food or quantity are recorded.

Examples:

- 1 XXL mouse
- 5 large Drosophila
- 1 micro cricket

The documented feeding is then included in the animal details and feeding history.

### 8.2 Feeding History

The feeding history displays an animal's previous feedings in chronological order.

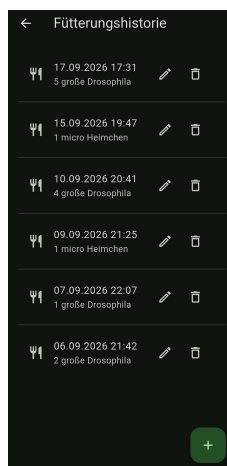
Each entry contains:

- Date
- Time
- Documented food or quantity

An existing entry can be corrected using the pencil icon.

An incorrect entry can be deleted using the trash icon.

Use + to add another feeding.



*The feeding history documents the time, food, and quantity and allows corrections to be made later.*

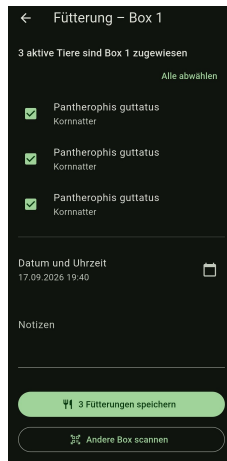
## 9. Quick Feeding via Enclosure Scan

In larger collections, animals do not have to be searched for individually in the animal list.

In the corresponding feeding workflow, the QR code of an enclosure is scanned first.

TerraManager then displays the active animals in that enclosure. One or more animals that were actually fed can be selected, and the feeding can be documented together.

A separate feeding entry is created for each selected animal.



*After scanning the enclosure, the animals that were actually fed can be selected directly.*

## Wrong Enclosure Scanned?

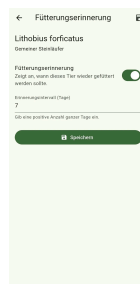
If the wrong QR code was scanned by mistake, Scan Another Enclosure returns to the scanner. It does not automatically scan the next enclosure; it lets you correct an accidental scan.

# 10. Feeding Reminders

Feeding reminders plan recurring feedings for each animal. You can use either a day interval or a fixed weekday schedule.

## 10.1 Set an Interval or Weekdays

Open the reminder action in the animal details and enable the feeding reminder. Choose either a positive whole number of days or fixed weekdays with a time and time zone. Recording a feeding does not shift the selected weekdays. Existing interval reminders remain compatible.



*Example: interval mode for an individual feeding reminder.*

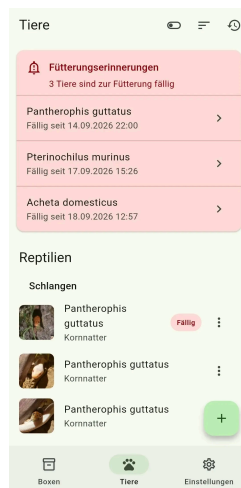
## 10.2 Next Feeding and Appointment List

TerraManager calculates the next feeding appointment from the selected plan and shows it in the animal details. The optional Next Feeding card in Animal Overview opens every active reminder in chronological order, whether the appointment is upcoming or already overdue.

## 10.3 When a Feeding Becomes Due

When the calculated appointment is reached or passed, TerraManager marks the animal as Due. The primary Animals navigation item and Animal Overview indicate active due feedings. The due list can be collapsed or expanded in both overview modes while its count remains visible.

The optional Next Feeding card also opens the complete chronological appointment list. Upcoming appointments therefore remain visible even when other animals are already overdue.



*Due feedings are highlighted centrally and on the respective animal entry.*

## Workflow

### Document feeding

- **apply an interval or fixed weekday plan**
- **display the next appointment**
- **place the appointment in chronological order**
- **mark the animal when feeding becomes due**
- **document the next feeding**

Intervals calculate the next appointment from feeding history. Fixed weekdays remain anchored to their calendar schedule.

## 11. Weight Management

Weight measurements can be documented for animals. Use Add Weight to save a new measurement. Previous measurements can be reviewed under Weight History.

→ calculate the next scheduled date

This allows animals with completely different feeding rhythms to be managed in parallel.

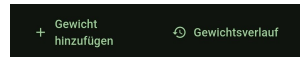
## 11. Weight Management

Weight measurements can be documented for animals.

Use **Add Weight** to save a new measurement.

Previous measurements can be reviewed under **Weight History**.

A new value does not simply replace the previous measurement. Weight development can therefore be documented over a longer period of time.



*Weight measurements can be stored continuously and reviewed through the weight history.*

## 12. Archiving Animals

An animal that is no longer part of the active collection does not have to be deleted.

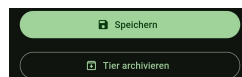
Use **Archive Animal** to move it to the animal history.

Additional information can be stored during archiving:

- Reason
- Archive date
- Archive note

This makes it possible, for example, to document why an animal left the active collection.

The existing animal information remains intact.



*Animals that are no longer active can be archived without losing their previous data.*

## 13. Animal History

The animal history contains archived animals separately from the active collection.

The overview may display:

- Image
- Animal name or species
- Archive reason
- Archive date



*Archived animals remain accessible through the animal history.*

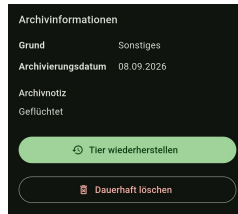
## 13.1 Archive Details

When an archived animal is opened, its stored information remains visible.

In addition, the **Archive Information** section displays:

- Reason
- Archive date
- Archive note

This ensures that it remains possible to determine later when and why an animal was archived.



*The reason, date, and optional notes permanently document the archiving event.*

## 13.2 Restore or Permanently Delete

Use **Restore Animal** to return an archived animal to the active collection.

Archiving is therefore reversible.

If the record should instead be removed permanently, **Delete Permanently** is available.

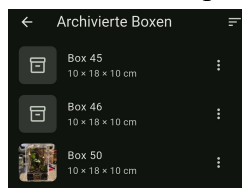
Note: For former animals whose history is still relevant, archiving should be used. **Delete Permanently**, on the other hand, permanently removes the record.

## 14. Archiving Enclosures

Enclosures that are no longer in use do not have to be deleted either.

They can be archived and removed from the active enclosure overview while their stored information is retained.

The **Archived Enclosures** view lists these containers separately from the active collection.



*Archived enclosures are stored separately from the active collection.*

When an archived enclosure is opened, information such as dimensions, enclosure ID, images, and other details remains available.

In addition, the archive reason, date, and archive note are displayed.

An archived enclosure can later be restored and returned to the active collection.

This is useful, for example, for temporarily stored terrariums or rearing containers.

# 15. Data Backup

As TerraManager is used over time, it contains not only current basic records but also images, feedings, weight histories, and historical information.

For this reason, functions are available to back up and restore the stored data.

## 15.1 Creating a Backup

The current TerraManager data set can be backed up using the backup function.

Creating a backup is particularly useful:

- Regularly during use
- Before major changes
- Before changing devices
- Before restoring older data

## 15.2 Restoring a Backup

An existing TerraManager backup can be selected and restored.

Because this changes the current state, TerraManager includes an additional protection mechanism.

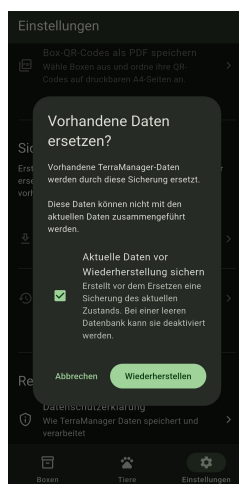
### Safety Backup

By default, a Safety Backup is provided before restoring data.

This creates an additional backup of the current state immediately before the actual restore process begins.

The Safety Backup is implemented as an opt-out feature: the additional protection is enabled by default but can be deliberately disabled by the user if required.

It is therefore not necessary to create an additional manual backup before every restore as long as the offered Safety Backup remains enabled.



*Before restoring data, TerraManager can automatically create a Safety Backup of the current state.*

# 16. A Typical TerraManager Workflow

TerraManager's features work together. A new collection can, for example, be set up and managed as follows:

## **01 — Create an Enclosure**

Record the terrarium or container, including dimensions and an image.

## **02 — Create an Animal**

Document the species, husbandry parameters, and any other known information.

## **03 — Link Animal and Enclosure**

Assign the animal to the corresponding enclosure.

## **04 — Generate a QR Code**

Export the enclosure code in the desired size and attach it to the container.

## **05 — Document Feeding**

Record the feeding directly for the animal or use the QR-based feeding workflow.

## **06 — Set a Feeding Schedule**

Choose either an interval or fixed weekdays with a time and time zone for each animal.

## **07 — Check Appointments and Reminders**

Review due animals and the chronological list of all next feeding appointments.

## **08 — Document Husbandry**

Continuously add images, weights, feedings, and other information.

## **09 — Preserve Historical Records**

Archive animals or enclosures that are no longer active instead of immediately deleting their data.

## **10 — Back Up Your Data**

Regularly create a backup of the growing data set.

## 17. TerraManager in Daily Use

After the initial setup, everyday use of TerraManager is largely reduced to a small number of recurring steps.

During routine care, an enclosure can be identified using its QR code. Feedings are assigned directly to the animals kept inside and stored in their individual histories.

At the same time, the animal overview shows whether additional animals are due for feeding. Individual intervals make it possible to manage, for example, a snake with a longer feeding cycle independently from a jumping spider or another animal that is fed more frequently.

Weights, images, and additional notes can be added whenever necessary. If an animal leaves the collection or an enclosure is taken out of service, the previous information remains available through the archive functions.

Over time, TerraManager therefore develops from a simple collection overview into an ongoing record of animal husbandry.

# 18. Shared Care in the Browser

## 18.1 A Separate Server Mode

Shared Care is an optional browser mode for several caregivers. Your own server on the local network owns the shared collection and local user accounts. Standalone Android and Web retain their separate databases; their records are not transferred or synchronized automatically. In Android Settings, the Shared care link opens the HTTPS address in an external browser. Sign in there with an account created by the server operator.

## 18.2 Overviews and Quick Actions

Boxes and Animals can be created, edited, archived and viewed in Big Picture Mode. Long-press or right-click a record for its context menu. Active records offer rename, edit, duplicate and archive; an Animal can also start a Feeding. Archived records can be restored from the menu. Permanent deletion is available only at the bottom of archived details and requires confirmation. An exclamation mark on the Animals tab indicates a due Feeding. Changes from other devices appear after the next refresh; review the new server state when an edit conflict is reported.

## 18.3 QR Workflows

Scan a Box QR label to open its details. In QR Feeding Mode, scan a Box and select the active Animals that were actually fed. When editing an Animal, a different active Box QR label can select its new Box; confirm the move before reassignment. Camera scanning needs trusted HTTPS and a browser with built-in QR recognition. If scanning is unavailable, Feeding Mode offers Choose Box, while Edit Animal retains the ordinary Box selector.

## 18.4 Export Box QR Codes

Settings - Box QR codes offers three actions: individual PNG files, one ZIP archive and a printable A4 PDF. Each export opens a checklist of active and archived Boxes; all are selected initially. For PDF, choose a physical code size between 6 and 20 mm. Files are generated in the browser and saved through its download facility.

## 18.5 Backups and Privacy

Administrators can download a portable backup of the shared collection and verify it in a separate test installation. The file does not contain server accounts; full recovery also requires the operator to back up the server data directory. Scanner camera frames are not uploaded. The browser sends a validated QR identifier only to the operator's server. The Android app needs no new network permission for its external browser link.

Setup, certificates, updates and field checks:

<https://codefrogcf.github.io/TerraManager/shared-care-deployment.html>

# **Addendum to Chapter 15: Password-Protected Backups**

## **15.3 Creating a protected backup**

When saving a portable backup, select “Protect with password”. Enter the password twice. The resulting .tmbackup file contains an encrypted archive. Keep the password separately from the file. TerraManager neither stores nor recovers it.

Saving a backup without a password remains possible. Older unprotected backups remain importable.

## **15.4 Restoring a protected backup**

Select the .tmbackup file as usual. TerraManager recognizes the protection and asks for the password before validating the backup. An incorrect password or a modified or damaged file prevents restore. You still confirm before the collection is replaced.

If the Safety Backup is enabled in standalone mode, that copy uses the same password. Cancelling its save dialog stops the restore.

## **Shared Care**

Administrators first save a current safety copy. A protected restore requires that copy to be password-protected too. Export encryption and import decryption take place in the browser. For restore, the operator's server receives the decrypted portable archive over the authenticated connection. This does not encrypt server databases or host-volume backups.